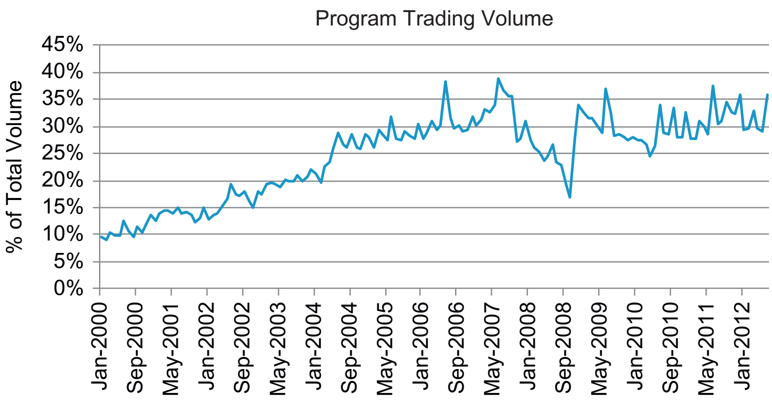

■ **Figure 1.8** Program Trading Volume.

bought or sold without any insight into the number of shares to transact. The recent increase in program trading activity is also due to large broker-dealers offering program trading at lower commission rates than traditional block executions. Technological improvements (e.g., Super DOT on the NYSE, Super Montage on NASDAQ, and in-house trading systems) have made it much easier for brokers to execute large lists of stocks more efficiently. Now, combine the large increase in program trading with the large reduction in block volume and we begin to appreciate the recent difficultly in executing large positions and the need for more efficient implementation mechanisms. Execution and management of program trades have become much more efficient through the use of trading algorithms, and these program trading groups have pushed for the advancement and improvement of trading algorithms as a means to increase productivity.

RECENT GROWTH IN ALGORITHMIC TRADING

To best position themselves to address the changing market environment, investors have turned to algorithmic trading. Since computers are more efficient at digesting large quantities of information and data, more adept at performing complex calculations, and better able to react quickly to changing market conditions, they are extremely well suited for real-time trading in today's challenging market climate. Algorithmic trading became popular in the early 2000s. By 2005, it accounted for about 25% of total volume. The industry faced an acceleration of algorithmic trading (as well as a proliferation of actual trading algorithms) where volumes increased threefold to 75% in 2009. The rapid increase in activity was largely due to the increased difficulty investors faced executing orders. During the financial crisis, it was not uncommon to see stock price swings of 5–10% during the